

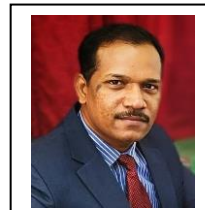


**Pimpri Chinchwad Education Trust's
Pimpri Chinchwad College of Engineering & Research
Ravet, Pune**



Academic Year: 2022-23

Faculty Profile



Name of Faculty:	Dr. Rahul Krishnaji Bawane		
Qualification :	Ph.D. - Mechanical Engineering, ME – Heat Power		
Designation :	Assistant Professor		
Department :	Mechanical Engineering		
Date of Joining :	23/12/2014		
Experience :	Teaching : 13 Years Industry: 4 Years		
Research Areas / Area of Specialization	Mechanical (Heat Power, IC Engine, Applied Thermodynamics)		
Courses / Subjects Taken:	1. Engineering Thermodynamics (2019 Pattern) 2. Applied Thermodynamics (2015 & 2019 Pattern) 3. Heat Transfer (2015 Pattern) 4. Basic Mechanical Engineering (2015 Pattern) 5. Engineering Graphics-I (2015 Pattern)		
IPR :	Title of Patent	Patent No.	Filed on
	1. Hinged And Extensible Car Flap To Shade Eyes From The Glare Of Sunlight	202121036946	15-08-2021
	2. Weight Sensor Operated Footrest	202121037009	16-08-2021
	3. Mechanical Ring Hose Fastener	202121037011	16-08-2021
	4. Air Pressure Operated Hose Fastener	202121036999	16-08-2021
	5. Taper Surface Drilling Fixture	202121036998	16-08-2021
	6. Adjustable Parallel Line Drawing Instrument	202121036997	16-08-2021
	7. Foot Pedals Force Conversion To Electrical Energy	202121036996	16-08-2021
	8. Paper Pinning Cum Punching Machine	202121036995	16-08-2021
	9. Pivoting Latch Pinning Stapler	202121036994	16-08-2021
	10. Color Coding For Tire Wear Rate	202121036992	16-08-2021
	11. Adjustable Hinged Lever Compass Drawing Instrument	202121036990	16-08-2021
	12. Mechanical Linkages Automatic Cleaning Of Ceiling Fan	202121037002	16-08-2021

	13. Headlight Upper-Dipper Automatic Shifter	202121036988	16-08-2021
	14. Fluid Cushioning Sport Shoe	202121036993	16-08-2021
	15. Fluid Pressure Operated Clutch Pen	202121036989	16-08-2021
	16. Automatic Extraction Guard For Bike Front Wheel To Arrest Mud/Water Spillage	202121036973	16-08-2021
	17. Waste Heat Recovery Of Vehicle Exhaust To Generate Plastic Fuel	202121036972	16-08-2021
	18. Pollution Under Control (PUC) Alert	202121036971	16-08-2021
	19. Automatic Adjustment Of Rear-View Mirrors	202121036970	16-08-2021
	20. Magnetic Force Anti-Gravity Ball Pen Refill	202121036969	16-08-2021
	21. Gesticulations Control Drone	202121036968	16-08-2021
	22. Offset Drilling Fixture For Cylindrical Object	202121036967	16-08-2021
	23. Spherical Ball Mandible Coupling	202121036966	16-08-2021
	24. Primacy Controlled Elevator Operation	202121036965	16-08-2021
	25. Automatic Shutdown System For Ignition System On Accident	202121036964	16-08-2021
	26. Ignition Switch Operated Foot Rest For Bike	202121036986	16-08-2021
	27. Mechanical Linkage Power Generating Shoes	202121036985	16-08-2021
	28. Conveyor Belt Driven Seedling Sowing Machine	202121036977	16-08-2021
	29. Caterpillar Tracks Drive For Off-Road Vehicles	202121036963	16-08-2021
	30. Bike Helmet Audibility Enhancement Air Breath Sliding Windows	202121036978	16-08-2021
	31. Automatic Synchronized Road Traffic Signals	202121036980	16-08-2021
	32. Sensitive Zone Automatic Flight Mode Shifter For Cell Phone	202121036959	16-08-2021
	33. Hot Air Assisted Energy Recovery Heating System	202121036958	16-08-2021
	34. Exhaust Gas Fuel Injection	202121036956	16-08-2021
	35. Exhaust Gas Assisted An Anti-Water Tire Splashes	202121036979	16-08-2021
	36. Fuel Dispensed Quantity Indicator On Vehicle	202121036961	15-08-2021
	37. Flexible Squeezer Toothpaste Dispenser	202121036953	15-08-2021
	38. Temperature Reducing Curtains	202121036951	15-08-2021
	39. Sliding Chin Guard Bike Helmet	202121036950	15-08-2021
	40. Bike Rider Weight Actuating Starting System	202121036955	16-08-2021
	41. PUC Dashboard Indicator	202121036987	16-08-2021
	42. Autonomous Sanitizer Mobil	202121036948	15-08-2021
	43. Mechanical Linkage Seedling Sowing Mechanism	202121036947	15-08-2021

44. Emergency Reserved Engine For Vehicle	202121036945	15-08-2021
45. Voice Command Gear Change Mechanism	202121036960	15-08-2021
46. Automatic Brake Speed Governor	202121036952	15-08-2021
47. Pathology Analysis Urinal System	202121036944	15-08-2021
48. Suction Cap Anti-Skid Footwear	202121036943	15-08-2021
49. Evaporation Cooling System For Car	202121036941	15-08-2021
50. Pre-Heating Fuel Injection Using Exhaust Gas	202121036975	16-08-2021
51. Broom Casing With Extensible Handle	202121036962	16-08-2021
52. Centrifugal Force Operated Ceiling Fan Blade Cleaning Attachment	201821049613	28-12-2018
53. Automatic Bike Passing High And Low Bean Signal	201821049611	28-12-2018
54. A Compass Attachment For Drawing Circle	201821049608	28-12-2018
55. Hand Held Stapler With Stapler Lock	201821049606	28-12-2018
56. Tire Wear Out Warning Visual Indicator	201821049603	28-12-2018
57. Copassenger Weight Sensing Automatic Opening And Closing Foot Rest Of Bike	201821049600	28-12-2018
58. Generation Of Electrical Energy Using Piezoelectric Cell On Bicycle Pedals	201821049594	28-12-2018
59. Ball Pen On/Off Using Fluid Pressure At Nozzle Body	201821049556	28-12-2018
60. Alteration In Paper Punching Machine To Staple Simultaneously	201821049567	28-12-2018
61. An Alteration In Ruler To Draw Parallel Line	201821049570	28-12-2018
62. Offset Drilling Fixture For Drilling Hole On Inclined Surface Of Any Object	201821049572	28-12-2018
63. Rotating Lock Ring And Fixed Slider Hose Clamp	201821049584	28-12-2018
64. Pneumatic Pressure Operated Hose Clamp	201821049586	28-12-2018
65. Extensible Sun Visor To Obstacle Sun Rays From The Centre Portion Of The Wind Shield Of Vehicle	201821049589	28-12-2018
66. Mechanical Linkage Operated Mudguard Extension For Two Wheeler Motor Vehicle	201821049590	28-12-2018
67. Non-Newtonian Fluid Shoe Cushioning And Shock Absorbing	201821049593	28-12-2018
68. Fuel Injection Assisted By Recirculation Of Exhaust Gas And Dilution Of Intake Charge	201821012914	05-04-2018
69. Production Of Plastic Pyro-Oil Using Waste Heat From Automobile Exhaust Gases	201821006521	21-02-2018
70. Standby Single Cylinder Engine Drive System For Car	201721040061	10-11-2017
71. Voice Recognized Gear Shifting For Automobile	201721038390	30-10-2017

72. Fuel Heating Before Injector Using Exhaust Gas For Diesel Engine	201721032362	13-09-2017
73. Speed Governing By Automatic Braking System For Automobiles	201721032364	13-09-2017
74. Customer Side Fuel Dispensed Measurement And Display System On Vehicle	201721026859	28-07-2017
75. Vacuum Water Evaporation And Humidification Air Cooling System For Car In Dry & Hot Climate	201721009979	23-03-2017
76. Pressurized Ink Ball Pen Refill Employing Magnetic Repulsion Force	201621029856	01-09-2016
77. Motorcycle Helmet Having Sliding Chin Guard With Helmet Locking To Mirror Rod	201621026956	08-08-2016
78. Pen To Push Out And Pull In Refill Automatically When Gripped And Un-Gripped Using Fluid Gripper	201621025803	28-07-2016
Title of Copyrights Registered (33 Registered and 131 Submitted)	Docket No./ Diary No.	Registered on
1. Laser Ignition System In An Internal Combustion Engine	L-83183-2019	03-04-2019
2. Automotive Thermoelectric Generator (ATEG) In IC Engine	L-82621-2019	03-04-2019
3. Production Of Biodiesel From Sesame Oil As An Alternative Fuel	L-83184-2019	03-04-2019
4. Peanut Oil Biodiesel Alternative For Diesel Fuel	L-82691-2019	02-04-2019
5. Olive Oil Biodiesel As An Alternative Fuel For Diesel Engine	L-82703-2019	02-04-2019
6. Alternative Energy Source Hydrogen Fuel Cell	L-82722-2019	02-04-2019
7. Transformation Of Waste Plastic Into Plastic Fuel	L-82623-2019	02-04-2019
8. Rice Bran Oil Edible Oil Feedstock For Biodiesel	L-83182-2019	02-04-2019
9. Propane As A Biodiesel An Alternative Source Of Energy	L-82724-2019	02-04-2019
10. Liquefied Natural Gas (LNG) As An Alternative Source Of Energy	L-82704-2019	01-04-2019
11. Advance Internal Combustion Engine Research	L-82705-2019	01-04-2019
12. Biodiesel From Mustard Oil As An Alternative To Diesel Fuel	L-82706-2019	01-04-2019
13. Biodiesel Production From Neem Oil	L-82035-2019	26-03-2019
14. Electric Energy As Alternative For Conventional Fuel	L-83157-2019	26-03-2019
15. Palm Oil Biodiesel Prospectus As An Alternative Fuel For Diesel	L-82226-2019	25-03-2019
16. Production Of Biodiesel From Animal Fats	L-83133-2019	23-03-2019
17. Acetylene As Alternative Fuel In Petrol Engine	L-83134-2019	23-03-2019
18. Ethanol As Alternative For Conventional Fuel	L-82912-2019	22-03-2019

	19. Biodiesel Production From Waste Cooking Oil	L-83131-2019	17-03-2019
	20. Acetylene As Alternative Fuel In Petrol Engine	L-82227-2019	17-03-2019
	21. Barley As A Bio-Fuel Crop	L-83246-2019	15-03-2019
	22. Working Phenomenon Of Six Stroke IC Engine	L-82512-2019	12-03-2019
	23. Coffee Bio fuel Feedstock For An Alternative Fuel	L-83258-2019	12-03-2019
	24. Biodiesel From Coconut Oil As An Alternative Fuel In IC Engines	L-82583-2019	12-03-2019
	25. Compressed Air As Alternate Fuel To Power The Vehicles	L-82065-2019	07-03-2019
	26. Coriander Seed Oil Methyl Esters As Biodiesel Fuel : Unique Fatty Acid Composition And Excellent Oxidative Stability	L-82073-2019	04-03-2019
	27. Rapeseed – A Potential Feedstock For Biodiesel	L-82064-2019	04-03-2019
	28. Positive Displacement Reciprocating And Rotary Compressor	L-79653-2018	14-11-2018
	29. IC Engine Systems	L-79654-2018	14-11-2018
	30. Testing Of IC Engine	L-79655-2018	14-11-2018
	31. Compression Ignition Engine	L-79656-2018	14-11-2018
	32. Spark Ignition Engine	L-79657-2018	14-11-2018
	33. Basic Of IC Engine	L-79658-2018	14-11-2018

Research Publications :

Journal Publications:	Year	Title of Paper	Published in
	2022	Diesel engine performance with nickel-oxide-doped Calophyllum oil biodiesel under varying injection timings, (9 th February, 2023).	Taylor & Francis, International Journal of Ambient Energy, Scopus Index Journal (Q3). doi.org/10.1080/01430750.2023.2172608
	2020	Impact analysis of emulsified Calophyllum oil biodiesel B100-WIC and B50-WIC on performance and emission characteristic of a diesel engine under variation in Compression Ratio	Taylor & Francis - International Journal of Ambient Energy (Scopus Indexed Journal ISSN 01430750) doi: 10.1080-01430750.2020.1852110
	2020	Impact Analysis of Calophyllum Inophyllum Oil Biodiesel on Performance and Emission Characteristic of Diesel Engine under Variation in Compression Ratio, Engine Load, and Blend Proportion	Taylor & Francis -International Journal of Ambient Energy (Scopus Indexed Journal ISSN 01430750). doi: 10.1080-01430750.2020.1730955
	2020	Indian energy scenario, biodiesel oxidation, impact on fuel system and engine performance & emission characteristics – a review	Manufacturing Technology Today (UGC Approved 3830, Peer Review Journal, ISSN 0972-7396).Vol. 19, No. 1, January 2020

	2019	Analysis of Impact of Use of Waste Cooking Oil Biodiesel on Engine Performance	Journal of Applied Science and Computations. (UGC Approved 41238, Peer Review Journal, ISSN 1076-5131). Volume VI, Issue II, February 2019
	2018	A Review on Mathematical Modeling in Biodiesel for IC Engines	Journal of Applied Science and Computations. (UGC Approved 41238, Peer Review Journal, ISSN 1076-5131). Volume 5, Issue 10, October 2018
	2018	Biofuel Indian Scenario, National Policies, Methodology, Opportunities and Challenges	Journal of Applied Science and Computations, (UGC Approved 41238, Peer Review Journal, ISSN 1076-5131). Volume 5, Issue 9, September-2018
	2018	Experimental Investigation of Performance Characteristics of CI Engine Fuelled With CalophyllumInnophyllum Oil Biodiesel Blend	International Journal of Advance Research in Science and Engineering. Vol 07, Issue no. 03, April 2018
Conference Publications:	Year	Title of Paper	Published in
	2023	Injector Deposition and Behavior Change of Diesel Engine Fueled with Calophyllum Oil Biodiesel Blend under 150 Hrs Endurance Tests.	(Accepted) SAE, WCX International Conference, Detroit, USA. 18-19 April 2023.
	2020	Characterization of Calophyllum oil Biodiesel – Alternative fuel to Diesel Engines	Techno-Societal 2020: 3rd International Conference on Advanced Technologies for Societal Applications. 11th – 12th December, 2020
	2020	A Review of Biodiesel Oxidation Stability and Chemistry	International Conference on Research Outlook, Innovations and Research Trends (ICROIRT - 2020) on 29-30 August 2020
	2019	Impact and Performance of Biodiesel on Emissions and Oxidation Stability – A Critical Review	International Conference on Advancements and Futuristic Trends in Mechanical and Materials Engineering. December 5-7, 2019. IIT Ropar
Books Publication:		1. Applied Thermodynamics (2015 Pattern) GigaTech Publication, ISBN 9789-3884-8009-3, Published in 2018. 2. Heat Transfer (2015 Pattern) GigaTech Publication, ISBN 9789-3884-8014-7, Published in 2019. 3. Applied Thermodynamics (2019 Pattern) Namya Publication, ISBN 978-93-91010-49-2, Published in 2021.	

Profile Link :	Orchid ID: https://orcid.org/0000-0001-9064-187X Google Scholar: Citations: 53, h-index: 4, i10-index: 2 https://scholar.google.com-citations?user=N7ZMuwAAAAJ&hl=en Personal Website: https://sites.google.com/pccoer.in/pccoer/home Blog: https://rahulbawane.blogspot.com/ YouTube Channel: https://www.youtube.com/channel/UCvRi6nB66JPG5xSivKn8Vtw/videos https://www.youtube.com/channel/UCXWLmHVLcsXNpbKf4u6NbVg
Product Developed:	Customized Bike for Indian Road* Solar Operated Desert Cooler*
Awards-Appreciation-Achievements:	1. World Records India; New record for 131 Copyrights in a Single Day on 13/03/2023. 2. World Records India; New record for 51 Patents in a Single Day on 16/08/2021. 3. Best scholar Award -2020, International Research Award on Science, Health and Engineering (SHEN)2020. 4. Total Paper Publications = 23 numbers; Citation = 53; h-index = 4; i-index = 2. 5. Total Patents Published; 80 numbers; Total Copyrights; 33 registered and 131 submitted. 6. Convener National Conference on IPR –CIPCIS 2019 and 2020. 7. Special Award for Contribution in Research and Development, PCCOER on 2018-19. 8. Certificate of Excellence & Innovativeness in teaching, PCCOER on 2021-22.
Membership:	1. QCFI Life Time Member ship (04-99-03-120481)
Industrial Training :	1. One Week Industrial Training, at Ascent Casting Technology, Chakan MIDC, Pune, 9 – 15 June 2018
Resource Person -Expert Session-Jury Invited:	1. One week FDP on Research Paper Writing and Publishing commencing from 20th to 24th February, 2023. (Resource Person for Session-3: Patent publication and copyright on 22 February 2.00 PM to 4.00 PM.) Nutan College of Engineering and Research, Talegaon Dabhade, Pune -410507. 22nd February, 2023. 2. Two-day Expert talk on IPR Awareness and Patent Drafting arranged by Department of Electronic and Computer Engineering on 13th and 14th September 2022. Modern College of Engineering, Shivajinagar, Pune – 411005. 13th - 14th September. 2022. 3. Intellectual Property Rights - Patents in Agriculture, 3 Days Webinar, MPKV Rahuri, 9-10 & 17 March 2021. 4. 3-Days Webinar on IPR-Patent, PCCOER Ravet, 10th – 12th December, 2020.

	- Guest Lecture on Isometric Projection, Nutan College of Engineering & research, Talegaon Dabhade, on 21st November 2019. - Performance Evaluation of IC Engine and Emerging Trends in Engine Performance Evaluation, Navsahyadri Group of Institute, FOE, Naigaon, on 6th March 2019 - Keynote Speaker : Two Days State Level workshop on 'Research Methodology, Indian Patent Drafting and Filing Procedure'. Imperical College of Engineering and Research, Wagholi, 11 & 12 -01-2019. - Guest Lecture on 'Advances in Reciprocating Air compressor and its Analysis', Nutan Maharashtra Vidya Polytechnic, Talegaon Dabhade, 16-03-2019. - Keynote Speaker : One Day Workshop on 'Idea Generation and Intellectual Property Rights (IPR)', Dr. D Y Patil School of Engineering, Lohegaon, 10-04-2019.
Workshop-FDP-STTP attended(more than a Week):	- Artificial Intelligence & Machine Learning Using Python, Finland Labs (A Unit of Revert Technology Pvt. Ltd.), 28th June – 09th July 2021. - Emerging trends in refrigeration & air-conditioning, PCCOER Ravet, 1-6 June 2020. - Research Techniques and Sustainable Development in Mechanical Engineering, St. Jonh College of Engineering and Management, Palghar, 29 June – 3 July 2020. - ICT Tools for Teaching, Learning Process and Institute, Electronics and ICT Academies held from 10th - 21st August, 2020. - Research Methodology and Techniques, Sandip Institute of Engineering and Management Nashik from 04/07/2020 to 06/07/2020. - Tools for Online Teaching Learning Process, Government Polytechnic, Jalgaon, 20th April – 17th May 2020. - Revolution in Engineering Education, School of Engineering & Technology, Sandip University, Nashik, 04th May 2020 To 08th May 2020. - Patenting for Innovations, Sanjivani College of Engineering Kopergaon, 25th – 29th May 2020. - Renewable Energy and Utilization, Sandip Institute of Engineering and Management Nashik, 26th – 30th May, 2020. - NPTEL online course (4 week : January - February 2019) on "Patent Drafting for Beginners" with a consolidated score of 73% (Elite). - NPTEL online course (12 week : January - April 2019) on "IC Engines and Gas Turbines" with a consolidated score of 63% (Elite). - NPTEL online course (8 week : August - September 2018) on "Introduction to Research" with a consolidated score of 81% (Elite) and 2 credit score. 2018. - NPTEL online course (12 week : July - October 2018) on "Intellectual Property" with a consolidated score of 82% (Elite).
Event Organized Inter department- Inter college- Other recognized Industry-IIT's-NPTEL-SPOKE N TUTORIAL- etc. :	- One-day workshop on "Research Fund Proposal, PCCOER Ravet, 28th January, 2023. - Three-Day Webinar on IPR, PCCOER Ravet, 10th & 12th December 2020.
Event Organized in Department:	Copyright session conducted for FE Div. E and F under CBS,

	submitted total 131 copyrights on 12 & 13 March, 2023. 2. Under IIC, IPR Cell, event coordinated for 51 Patent Filed in a 24 Hours on August 15, 2021.
University level appointments:	1. Syllabus Formation Team Member for Applied Thermodynamics (2019 Pattern). 2. Paper Setter for Applied Thermodynamics (2019 Pattern). 3. Paper Setter for Heat Transfer (2015 Pattern).
Administrative Duties:	1. Vice President Institute Innovation Council (IIC). 2. IPR Coordinator in IIC. 3. Innovation, Patent and Copyright Coordinator. 4. Research and Development Cell Coordinator. 5. NBA Criteria-5 Coordinator. 6. NAAC Criteria-3 Coordinator.
Academics Duties:	1. Mentor SE Mechanical Batch B1 2022-22. 2. Internal Exam Coordinator – Dept. of Mechanical Engg.
Any Other	1. Revenue generated Rs. 98,000/- through PCCOER Patent Consultancy.